

Hyeonji Kim

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Research Interests

Keywords: Natural Language Processing; Database Query Processing; Deep Learning; Graph Databases; Graph Data Analysis; Big Data Processing.

- LLM-driven data systems & agents: dataset curation, augmentation, and distillation for domain adaptation and fine-tuning; tool-augmented agents for grounded reasoning and action.
- Data systems for ML: reproducible data/feature pipelines, end-to-end measurement, and model-in-the-loop monitoring.
- Large-scale retrieval & indexing: hierarchical and multi-vector embeddings, cross-modal (text-image) retrieval, time-aware search, and approximate nearest neighbor (ANN) methods.
- Natural-language interfaces to databases: robust dataset augmentation/cleaning and principled evaluation for NL2SQL, and multi-turn dialogue system to databases
- Event-centric intelligence: detecting and modeling real-world events from news, social media, and enterprise data for downstream analytics and decision support.

Education

Ph.D., Convergence IT Engineering (Integrated M.S.–Ph.D.), Pohang University of Science and Technology (POSTECH), Aug 2024.

Dissertation: “Systematic Data Augmentation and Cleaning Techniques to Improve Text-to-SQL Models.” Advisor: Prof. Wook-Shin Han.

B.S., Creative IT Engineering (Magna Cum Laude), Pohang University of Science and Technology (POSTECH), Feb 2016.

Appointments

Postdoctoral Fellow, Database & Data Mining Lab (DSLAB), POSTECH, Aug 2024 – Present.

Full-time Research Intern, Oracle Labs (PGX team), CA, Dec 2015 – Mar 2016.

Research Projects (Selected)

Methodology for Extracting Notable Stocks from Domestic Economic News Using Generative AI — Project Lead, Apr 2025 – Present.

High-level pipeline for event detection and stock candidate surfacing (research prototype). Responsibilities: project lead, problem formulation, evaluation design, and coordination with industry partner.

SW StarLab “Development of a Conversational and Self-Tuning DBMS” — Project Lead, 2018 – Present.

Led efforts to make DBMSs usable to non-experts by building a natural-language interface to databases and automatic physical/knob tuning that simplifies performance optimization for practitioners. Responsibilities included NL2SQL modeling, dataset curation/augmentation, and evaluation infrastructure for robust QA over RDBMS.

Smart Structuring of Large-Scale Dark Data — Project Lead, 2017.

Developed a machine-learning-based framework to convert unstructured enterprise “Dark Data” into structured data. Delivered workflow components (preprocessing, candidate-set mapping, feature extraction, supervised learning and inference), a pilot on public job announcements, and an end-to-end automation plan including SQL/Python APIs.

Publications (Selected)

Kyoungmin Kim, [Hyeonji Kim](#), George Fletcher, Wook-Shin Han. “Combining Sampling and Synopses with Worst-Case Optimal Runtime and Quality Guarantees for Graph Pattern Cardinality Estimation.” In Proc. ACM SIGMOD, 2021.

[Hyeonji Kim](#), Byeong-Hoon So, Wook-Shin Han, Hongrae Lee. “Natural Language to SQL: Where Are We Today?” Proc. VLDB 13(10):1737–1750, 2020.

[Hyeonji Kim](#), Juneyoung Lee, Sourav S. Bhowmick, Wook-Shin Han, JeongHoon Lee, Seongyun Ko, Moath H. A. Jarrah. “DUALSIM: Parallel Subgraph Enumeration in a Massive Graph on a Single Machine.” In Proc. ACM SIGMOD, 2016.

International Patents

Wook-Shin Han, Hyuk-Kyu Kang, Hyeonji Kim. “Method for Enhancing Learning Data Set in Natural Language Processing System.” U.S. App. No. 18/725,248, June 2024.

Wook-Shin Han, Hyuk-Kyu Kang, Hyeonji Kim. “Apparatus and Method for Processing Natural Language Query about Relational Database Using Transformer Neural Network.” U.S. App. No. 17/964,807, Oct 2022.

Wook-Shin Han, In-Hyuk Na, Hyeonji Kim, Hyuk-Kyu Kang. “Column and Table Prediction Method for Text-to-SQL Query Translation Based on a Neural Network.” PCT/KR2019/018491, Dec 2019.

Honors & Scholarships

Google Anita Borg Scholars APAC, 2016.

POSTECH–SUNY Korea IT Consilience Creative Program (ITCCP) Scholarship, 2012–2020.

Teaching & Mentoring

Teaching Assistant / Instructor Support (Samsung Electronics): Data Foundation (Oct 2017), Advanced Data Expert Course (Jul–Nov 2019), Data Programming Course (Apr 2020).

Reviewing Service

ACM SIGMOD (2016, 2018, 2021–2022, 2026); VLDB (2016, 2018–2021, 2025); KDD (2017–2021); ICDE (2020–2021); EDBT (2018); IEEE TKDE (2016).

Languages

Korean (Native); English (Professional).

Programming: Python (Expert); C++ (Expert); SQL (Expert); Java (Advanced); JavaScript (Working proficiency); R (Working proficiency); CUDA C/C++ (Working proficiency)